

Home Credit Default Risk – Machine Learning Project

Problem Statement

Financial institutions often reject loan applications from individuals with limited credit history, even if they are reliable borrowers. The goal of this project is to develop a predictive model that estimates the probability of a client defaulting on a loan, allowing lenders to make fairer and data-driven decisions.

Approach

The project involved integrating multiple datasets from the Home Credit database, including applicant information, previous loans, credit card balances, and payment histories. Data preprocessing included cleaning missing values, encoding categorical features, scaling numerical data, and balancing class distributions. Several supervised learning models were trained and compared based on performance metrics.

Implementation Overview

Feature aggregation was performed by merging auxiliary datasets on the applicant ID, computing statistical measures such as mean and max for numeric fields. After preprocessing, six models were evaluated: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, SVM, and Multi-Layer Perceptron. Logistic Regression achieved the best performance with an accuracy of 69.97% and ROC-AUC of 0.7629. The pipeline also produced feature importance visualizations and saved comparative model metrics.

Challenges and Conclusions

Major challenges included handling large datasets with millions of records, optimizing memory usage, and addressing class imbalance. Logistic Regression was found to be the most stable and interpretable model. The analysis revealed that features like external credit scores (EXT_SOURCE_2, EXT_SOURCE_3) and age were the strongest predictors of default risk. This project demonstrates how machine learning can enhance credit risk assessment and promote financial inclusion.